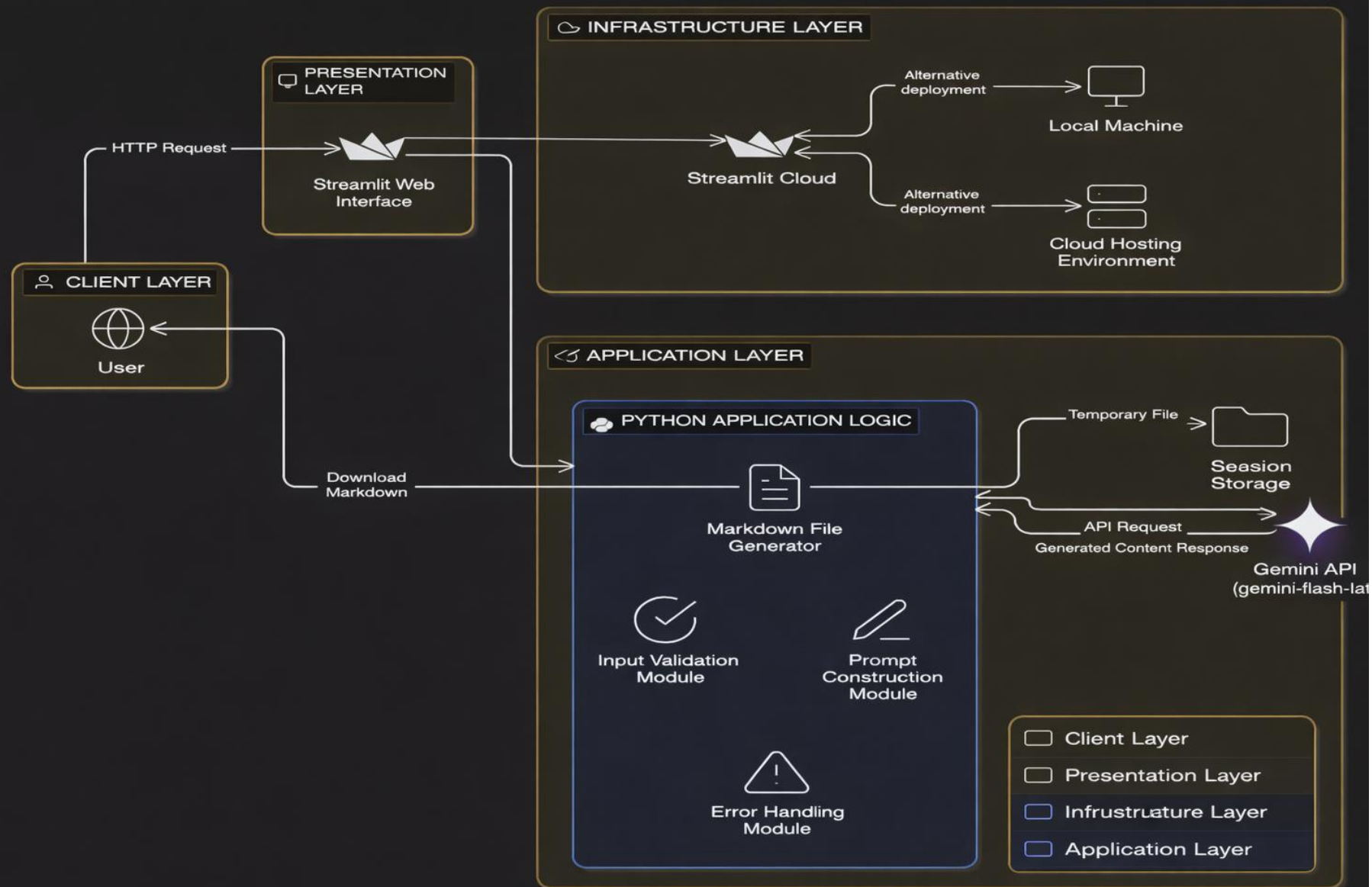


**Project Design Phase-II**  
**Technology Stack (Architecture & Stack)**

|               |                                  |
|---------------|----------------------------------|
| Date          | 6 February 2026                  |
| Team ID       | LTVIP2026TMIDS57703              |
| Project Name  | Civil Engineering Insight Studio |
| Maximum Marks | 4 Marks                          |

**Technical Architecture:**

The Deliverable includes the architectural diagram as below and the information as per the table1 & table 2



**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                                                                                               | Technology                                                          |
|------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1.   | User Interface                  | Web-based interface where users upload construction images, enter analysis request, and view generated structural report. | Streamlit (Python-based Web UI)                                     |
| 2.   | Application Logic-1             | Handles image upload, user input validation, and workflow control.                                                        | Python                                                              |
| 3.   | Application Logic-2             | Prompt construction logic for structured civil engineering analysis.                                                      | Python (Prompt Engineering Logic)                                   |
| 4.   | Application Logic-3             | AI-powered structural analysis via external model integration.                                                            | Google Gemini API (gemini-flash-latest)                             |
| 5.   | Database                        | No persistent database used (stateless application).                                                                      | Not Applicable                                                      |
| 6.   | Cloud Database                  | Not used in current implementation                                                                                        | Not Applicable                                                      |
| 7.   | File Storage                    | Temporary storage of generated report during session runtime.                                                             | Local Runtime (Streamlit Session State)                             |
| 8.   | External API-1                  | AI-based multimodal analysis (Image + Text).                                                                              | Google Gemini API                                                   |
| 9.   | External API-2                  | Not used                                                                                                                  | Not Applicable                                                      |
| 10.  | Machine Learning Model          | Large Multimodal Language Model for structural description and analysis.                                                  | Gemini Vision / Gemini Flash Model                                  |
| 11.  | Infrastructure (Server / Cloud) | Application can be deployed locally or on cloud hosting platforms.                                                        | Local System / Streamlit Cloud / Python-supported Cloud Environment |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                                             | Technology                               |
|------|--------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1.   | Open-Source Frameworks   | Framework used to build and deploy the web interface                                                    | Streamlit (Open-source Python framework) |
| 2.   | Security Implementations | API key secured using environment variables (.env) or Streamlit secrets; no sensitive user data stored. | python-dotenv / Streamlit Secrets        |
| 3.   | Scalable Architecture    | Stateless architecture; scalability depends on hosting environment and Gemini API limits.               | Streamlit + Cloud Hosting                |
| 4.   | Availability             | Application availability depends on hosting platform uptime.                                            | Streamlit Cloud / Cloud Hosting Provider |
| 5.   | Performance              | AI analysis response time depends on Gemini API latency; lightweight UI ensures minimal frontend delay. | Python + Gemini API                      |